

Employment

Product Engineer Intern	Isomer AI	May 2025 – Present
--------------------------------	------------------	---------------------------

- Ship 10+ production AI features end-to-end: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows, and REST API integrations; 394 commits across 7 production deployments.
- Own backend stack (PostgreSQL, Redis, Auth0 RBAC, Docker, Terraform, Sentry); debug and resolve production incidents.

Graduate Research Assistant	NC State University	April 2025 – Present
------------------------------------	----------------------------	-----------------------------

- Deploy cloud genomic platform on GCP (Cloud Run) with JBrowse 2 genome browser; support 100+ researchers, 500GB+ datasets.
- Improve ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.

Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
----------------------------------	-----------------------------	----------------------------

- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved processing accuracy by 25%.
- Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.

AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
---------------------	------------------	----------------------------

- Led end-to-end SDLC for data-driven system automating support ticket analysis, reducing processing time by 15%.
- Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
--------------------	--	----------------------------

- Master of Science in Computer Science
- Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.

Mumbai, India	Mumbai University	Aug 2020 – May 2024
----------------------	--------------------------	----------------------------

- Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.

Projects

VectorLift	Semantic Search Engine	2026
-------------------	-------------------------------	-------------

- Built semantic search engine in Python on MS MARCO: BM25 sparse retrieval, dense bi-encoder with in-batch negatives, cross-encoder re-ranker; Kafka-driven Qdrant corpus indexing pipeline.
- Scored NDCG@10, MRR@10, MAP; built FastAPI serving with per-stage latency tracking and Streamlit dashboard.

KubeServe	Kubernetes LLM Inference Platform	2026
------------------	--	-------------

- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing.
- Instrumented OTel + Prometheus + Grafana observability; Helm + ArgoCD GitOps; k6 to 500 VUs; 88 pytest tests.

LLM Eval Harness	Independent Project	2025
-------------------------	----------------------------	-------------

- Built vendor-neutral LLM benchmark comparing Claude, GPT-4o, Gemini across 2,000 frozen examples and 3 task types in Python.
- Tracked hallucination rate, p50/p95 latency, cost-per-call; applied bootstrap CIs and McNemar tests across 6 metrics.

AlgoPulse	Agentic AI Platform	2025
------------------	----------------------------	-------------

- Built agentic AI orchestration platform on Temporal.io and Kafka supporting multi-step reasoning pipelines; sustained 2,000 async inference requests per second under sustained production load.
- Replaced API polling with Supabase Edge Functions for real-time AI streaming; eliminated state loss under peak traffic.

Languages and Technologies

- **Languages:** Python, TypeScript, JavaScript, Java, SQL, Bash
- **Data & Pipelines:** Kafka, Spark, Hadoop, Temporal.io, PostgreSQL, Redis, Qdrant, REST APIs, Docker
- **Cloud & DevOps:** AWS, GCP, Kubernetes, Terraform, GitHub Actions, Grafana, OpenTelemetry, Prometheus
- **AI/ML:** Claude/GPT-4o/Gemini APIs, LangChain, RAG, LLM evaluation, prompt engineering

Additional Experience and Awards

- **IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- **National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- **Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- **Technical Writing:** Published 3 technical deep-dives on LLM model collapse, AI agent security, and neural audio architecture at kulkarniaditya.com/blog